Expt No: **Date:**

DESIGN OF IIR FILTERS

Aim: To design Butterworth low pass filter satisfying the constraints

$0.707 \leq |H(e^{jw})| \leq 1$ for $0 < < \pi/2$

$|H(e^{jw})| \leq 0.2$ for $3\pi/4 < < \pi$

With T=1 sec using (a) bilinear transformation (b) impulse invariance method. Plot the magnitude response of the filter and compare the two methods by means of order and filter coefficients.

Software Used: MATLAB s/w

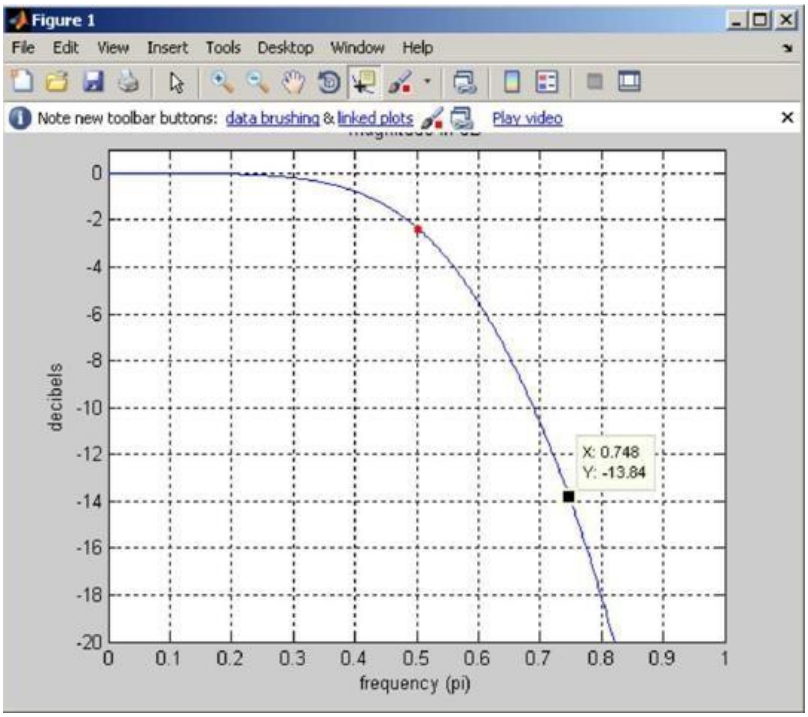
MATLAB Code:

% Bilinear transformation:

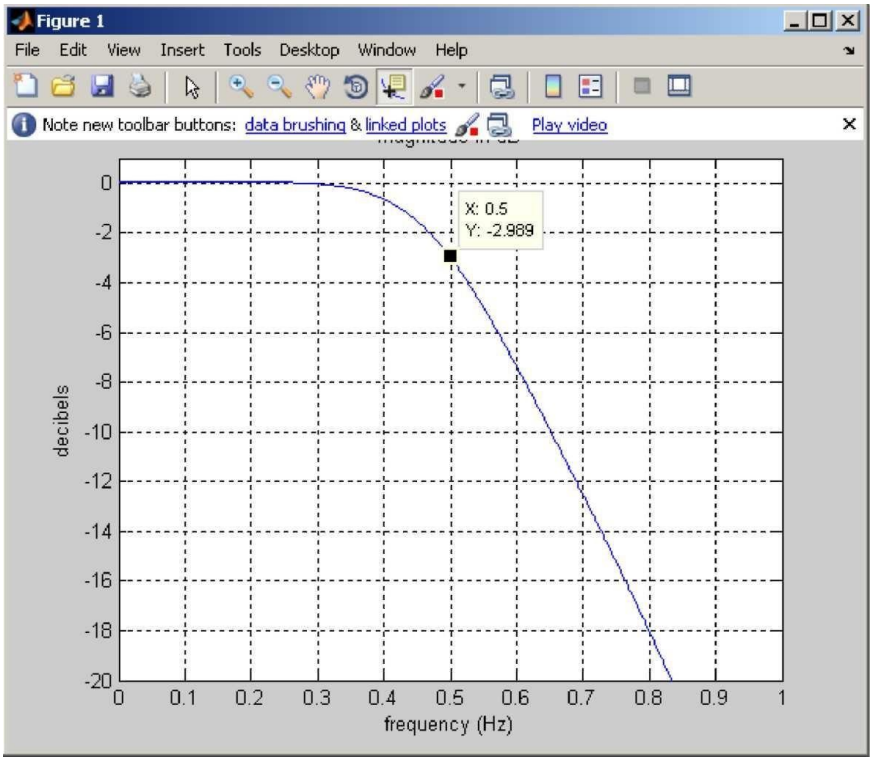
```
clc;
clear all;
close all;
wp = 0.5*pi; % digital passband frequency in Hz (normalized)
ws = 0.75*pi; % digital stopband frequency in Hz (normalized)
Rp=20*log10(1/sqrt(2)); % passband ripple in dB
Rp=-Rp;
As = 20*log10(0.2); % stopband attenuation in dB
As=-As;
% Analog Prototype Specifications:
Fs = 1; T = 1/Fs;
OmegaP = (2/T)*tan(wp/2); % prewarp prototype passband frequency
OmegaS = (2/T)*tan(ws/2); % prewarp prototype stopband frequency
% Analog butterworth Prototype Filter Calculation:
[n omegac]=buttord(OmegaP, OmegaS, Rp, As,'s');
[c d] = butter(n,omegac,'s');
% Bilinear Transformation:
[b, a] = bilinear(c, d, Fs);
 [h db] = freqz(b,a); plot(db/pi,20*log10(abs(h)));
 Axis ([0 1 -20 1]) ;
grid;
xlabel('frequency (Hz)');
 ylabel('decibels');
 title('Magnitude in dB');
```

DESIGN OF IIR LOWPASS FILTER

1. USING BILINEAR TRANSFORMATION METHOD



2. USING IMPULSE INVARIANT METHOD



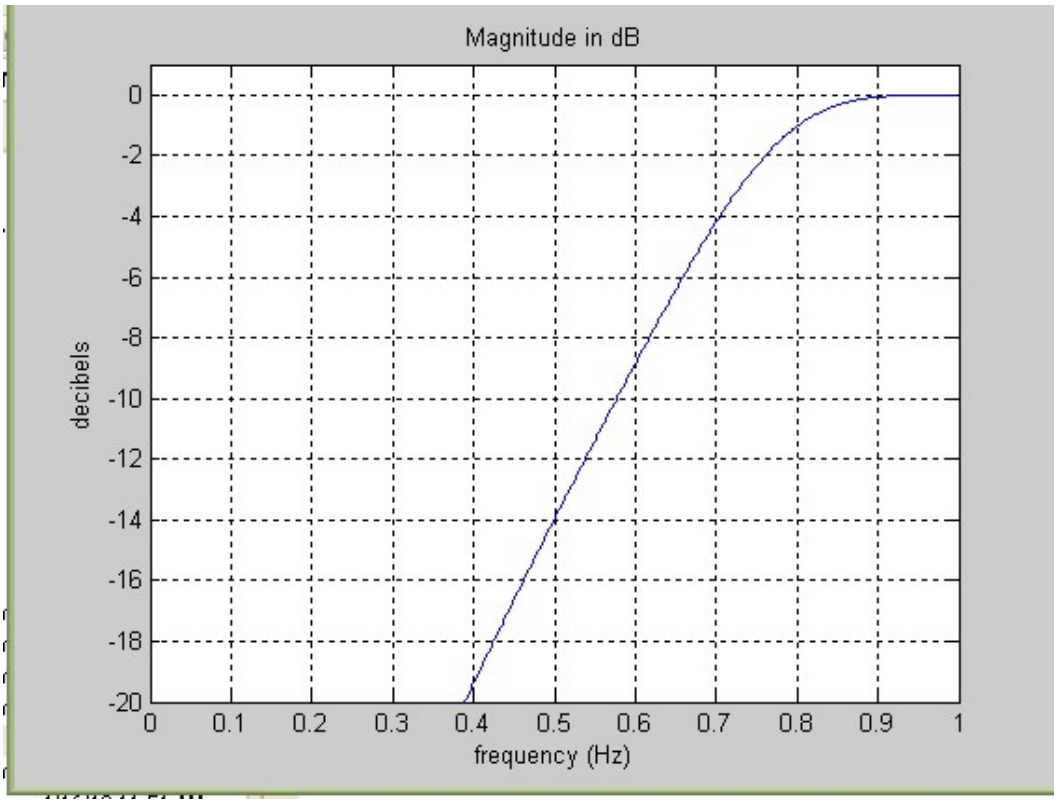
%IMPULSE INVARIANT METHOD

```
clc;
clear all;
close all;
wp = 0.5*pi; % digital passband frequency in Hz (normalized)
 ws = 0.75*pi; % digital stopband frequency in Hz (normalized)
 Rp=20*log10(1/sqrt(2)); % passband ripple in dB
Rp=-Rp;
As = 20*log10(0.2); % stopband attenuation in dB
As=-As;
% Analog Prototype Specifications:
Fs = 1; T = 1/Fs;
OmegaP = wp/T; % prewarp prototype passband frequency
OmegaS = ws/T; % prewarp prototype stopband frequency
% Analog butterworth Prototype Filter Calculation:
[n omegac]=buttord(OmegaP, OmegaS, Rp, As,'s');
[c d] = butter(n,omegac,'s');
% Impulse Invariant
Transformation:
[b, a] = impinvar(c, d, Fs);

[h db] = freqz(b,a); plot(db/pi,20*log10(abs(h)));
Axis ([0 1 -20 1])
grid
xlabel('frequency (Hz)');
ylabel('decibels');
title('Magnitude in dB');
```

DESIGN OF IIR HIGH PASS FILTER

1. USING BILINEAR TRANSFORMATION METHOD



IIR FILTER (HIGH PASS)

Aim: To design Butterworth high pass filter satisfying the constraints

$$|H(e^{jw})| \leq 0.2 \quad \text{for } 0 < < \pi/2$$

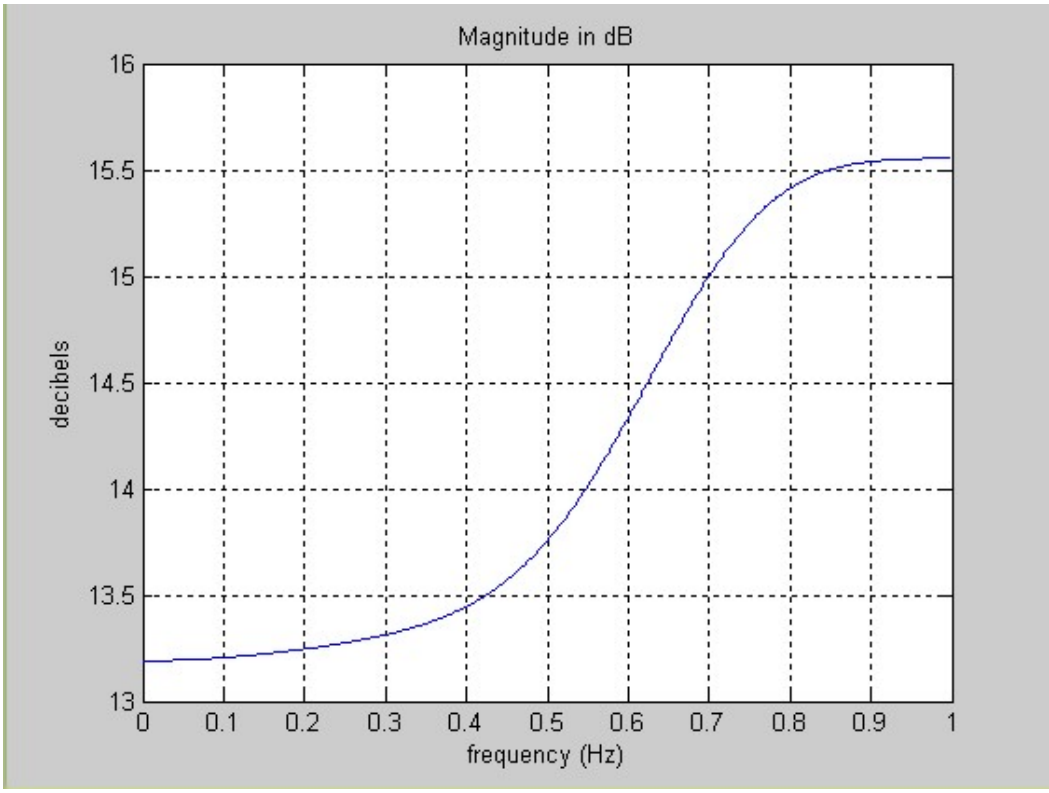
$$0.707 \leq |H(e^{jw})| \leq 1 \quad \text{for } 3\pi/4 < < \pi$$

With T=1 sec using (a) bilinear transformation (b) impulse invariance method. Plot the magnitude response of the filter and compare the two methods by means of order and filter coefficients.

% Bilinear transformation:

```
clc;;
clear all;
close all;
wp = 0.75*pi; % digital passband frequency in Hz (normalized)
ws = 0.5*pi; % digital stopband frequency in Hz (normalized)
Rp=20*log10(1/sqrt(2)); % passband ripple in dB
Rp=-Rp;
As = 20*log10(0.2); % stopband attenuation in dB
As=-As;
% Analog Prototype Specifications:
Fs = 1;
 T = 1/Fs;
OmegaP = (2/T)*tan(wp/2); % prewarp prototype passband frequency
OmegaS = (2/T)*tan(ws/2); % prewarp prototype stopband frequency
% Analog butterworth Prototype Filter Calculation:
[n omegac]=buttord(OmegaP, OmegaS, Rp, As,'s');
[c d] = butter(n,omegac,'high','s');
% Bilinear Transformation:
[b, a] = bilinear(c, d, Fs);
 [h db] = freqz(b,a);
plot(db/pi,20*log10(abs(h)));
axis([0 1 -20 1])
grid
xlabel('frequency (Hz)');
ylabel('decibels');
title('Magnitude in dB');
```

2. USING IMPULSE INVARIANT METHOD



%IMPULSE INVARIENCE METHOD

```
clc;
clear all;
close all;
% digital passband frequency in Hz (normalized)
wp = 0.75*pi;
% digital stopband frequency in Hz (normalized)
ws = 0.5*pi;
% passband ripple in dB
Rp=20*log10(1/sqrt(2));
Rp=-Rp;
% stopband attenuation in dB
As = 20*log10(0.2);
As=-As;
% Analog Prototype Specifications:
Fs=1;
T = 1/Fs;
% prewarp prototype passband frequency
OmegaP = wp/T;
% prewarp prototype stopband frequency
OmegaS = ws/T;
% Analog butterworth Prototype Filter Calculation:
[n omegac] =buttord (OmegaP, OmegaS, Rp, As, 's');
[c d] = butter (n, omegac, 'high', 's');
% Impulse Invariant Transformation:
[b, a] = impinvar (c, d, Fs);
[h db] = freqz (b, a);
plot (db/pi, 20*log10(abs(h)));
axis ([0 1 -20 1]); grid
xlabel ('frequency (Hz)');
ylabel ('decibels');
title ('Magnitude in dB');
```

Expt No: **Date:**

DESIGN OF IIR FILTERS

Aim: To design Chebyshev type-1 low pass filter satisfying the constraints

$0.707 \leq |H(e^{jw})| \leq 1$ for $0 < < \pi/2$

$|H(e^{jw})| \leq 0.2$ for $3\pi/4 < < \pi$

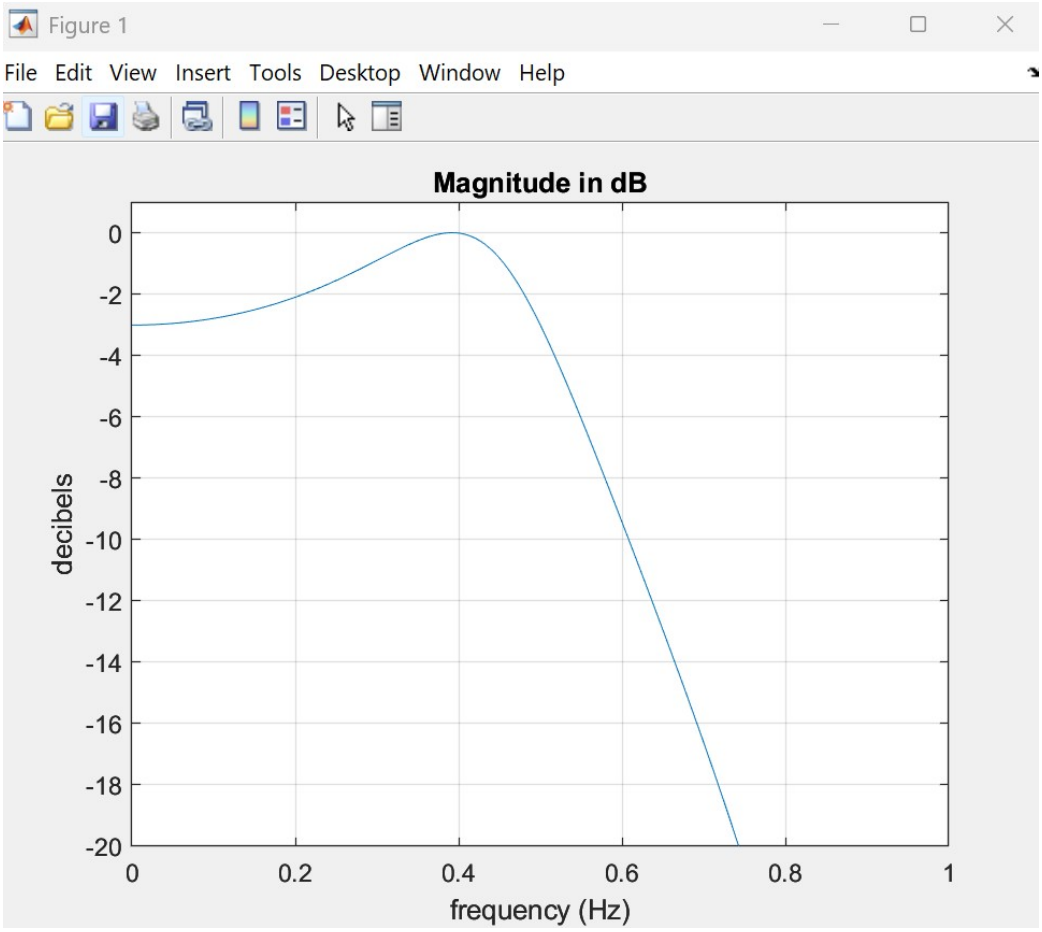
With T=1 sec using (a) bilinear transformation (b) impulse invariance method. Plot the magnitude response of the filter and compare the two methods by means of order and filter coefficients.

Software Used: MATLAB s/w

MATLAB Code:

% Bilinear transformation:

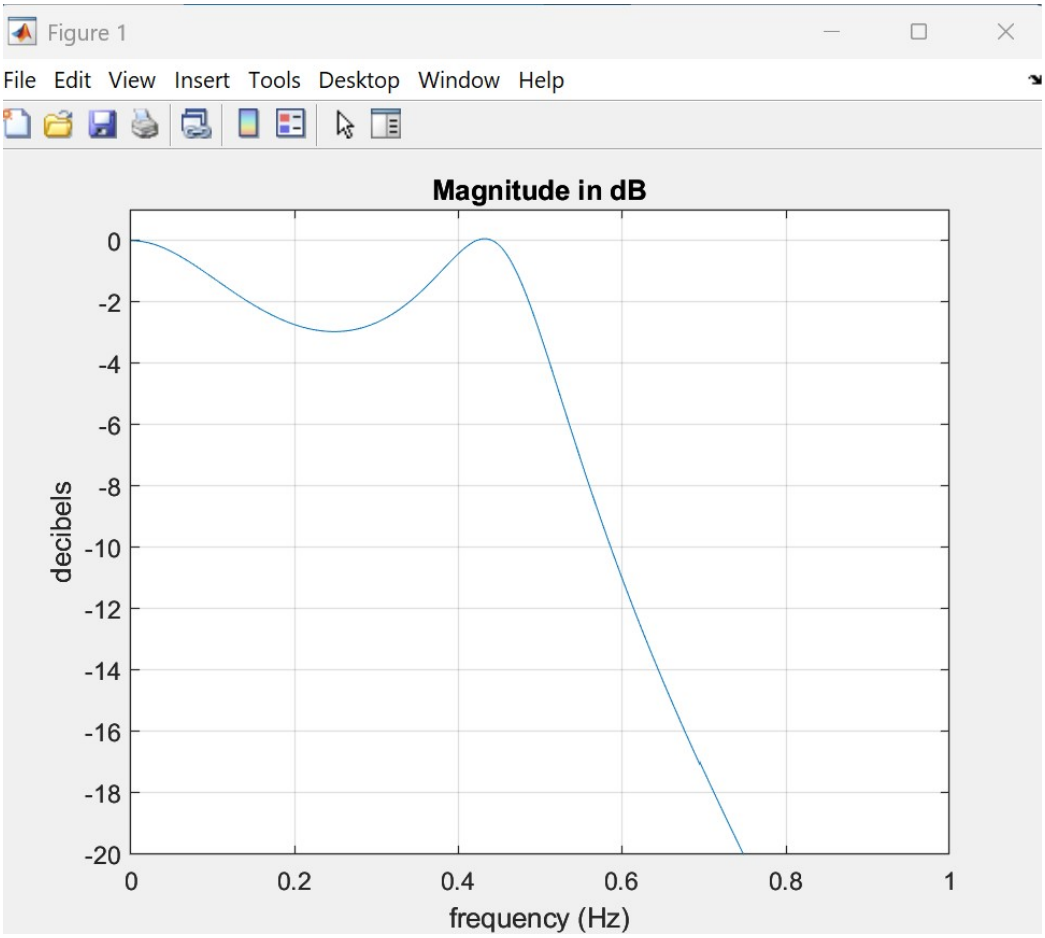
```
clc;
clear all;
close all;
wp = 0.5*pi; % digital passband frequency in Hz (normalized)
ws = 0.75*pi; % digital stopband frequency in Hz (normalized)
Rp=20*log10(1/sqrt(2)); % passband ripple in dB
Rp=-Rp;
As = 20*log10(0.2); % stopband attenuation in dB
As=-As;
% Analog Prototype Specifications:
Fs = 1; T = 1/Fs;
OmegaP = (2/T)*tan(wp/2); % prewarp prototype passband frequency
OmegaS = (2/T)*tan(ws/2); % prewarp prototype stopband frequency
% Analog butterworth Prototype Filter Calculation:
[n omegac] =cheb1ord (OmegaP, OmegaS, Rp, As, 's');
[c d] = cheby1(n, Rp, omegac,'s');
% Bilinear Transformation:
[b, a] = bilinear(c, d, Fs);
 [h db] = freqz(b,a); plot(db/pi,20*log10(abs(h)));
 axis ([0 1 -20 1]) ;
grid;
xlabel('frequency (Hz)');
 ylabel('decibels');
 title('Magnitude in dB');
```


Figure 1
File Edit View Insert Tools Desktop Window Help
Magnitude in dB
decibels
0
-2
-4
-6
-8
-10
-12
-14
-16
-18
-20
0
0.2
0.4
0.6
0.8
1
frequency (Hz)

%IMPULSE INVARIANT METHOD

```
clc;
clear all;
close all;
wp = 0.5*pi; % digital passband frequency in Hz (normalized)
 ws = 0.75*pi; % digital stopband frequency in Hz (normalized)
 Rp=20*log10(1/sqrt(2)); % passband ripple in dB
Rp=-Rp;
As = 20*log10(0.2); % stopband attenuation in dB
As=-As;
% Analog Prototype Specifications:
Fs = 1; T = 1/Fs;
OmegaP = wp/T; % prewarp prototype passband frequency
OmegaS = ws/T; % prewarp prototype stopband frequency
% Analog butterworth Prototype Filter Calculation:
[n omegac] =cheb1ord (OmegaP, OmegaS, Rp, As, 's');
[c d] = cheby1 (n,Rp, omegac,'s');
% Impulse Invariant
Transformation:
[b, a] = impinvar(c, d, Fs);

[h db] = freqz(b,a); plot(db/pi,20*log10(abs(h)));
axis ([0 1 -20 1])
grid
xlabel('frequency (Hz)');
ylabel('decibels');
title('Magnitude in dB');
```


Figure 1
File Edit View Insert Tools Desktop Window Help
Magnitude in dB
decibels
0
-2
-4
-6
-8
-10
-12
-14
-16
-18
-20
0
0.2
0.4
0.6
0.8
1
frequency (Hz)

IIR FILTER (HIGH PASS)

Aim: To design Chebyshev Tyoe 1 high pass filter satisfying the constraints

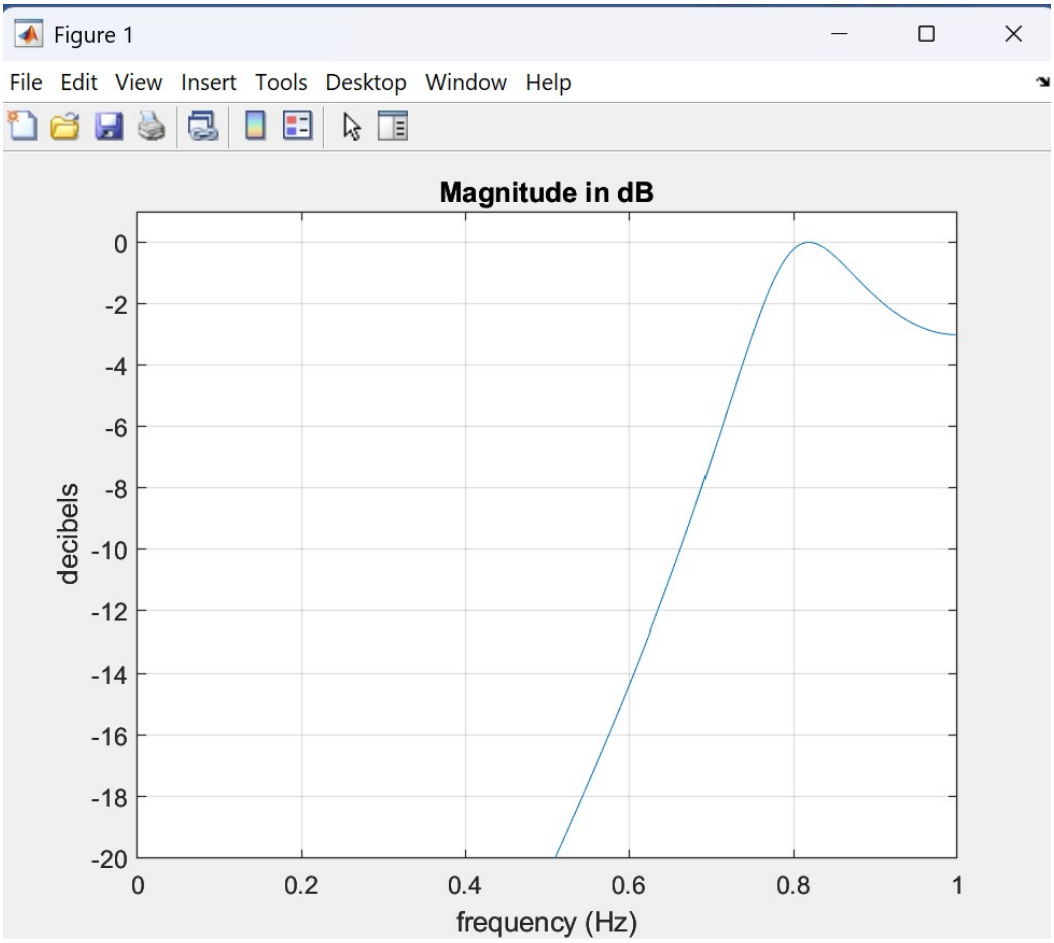
$|H(e^{jw})| \leq 0.2$ for $0 < < \pi/2$

$0.707 \leq |H(e^{jw})| \leq 1$ for $3\pi/4 < < \pi$

With T=1 sec using (a) bilinear transformation (b) impulse invariance method. Plot the magnitude response of the filter and compare the two methods by means of order and filter coefficients.

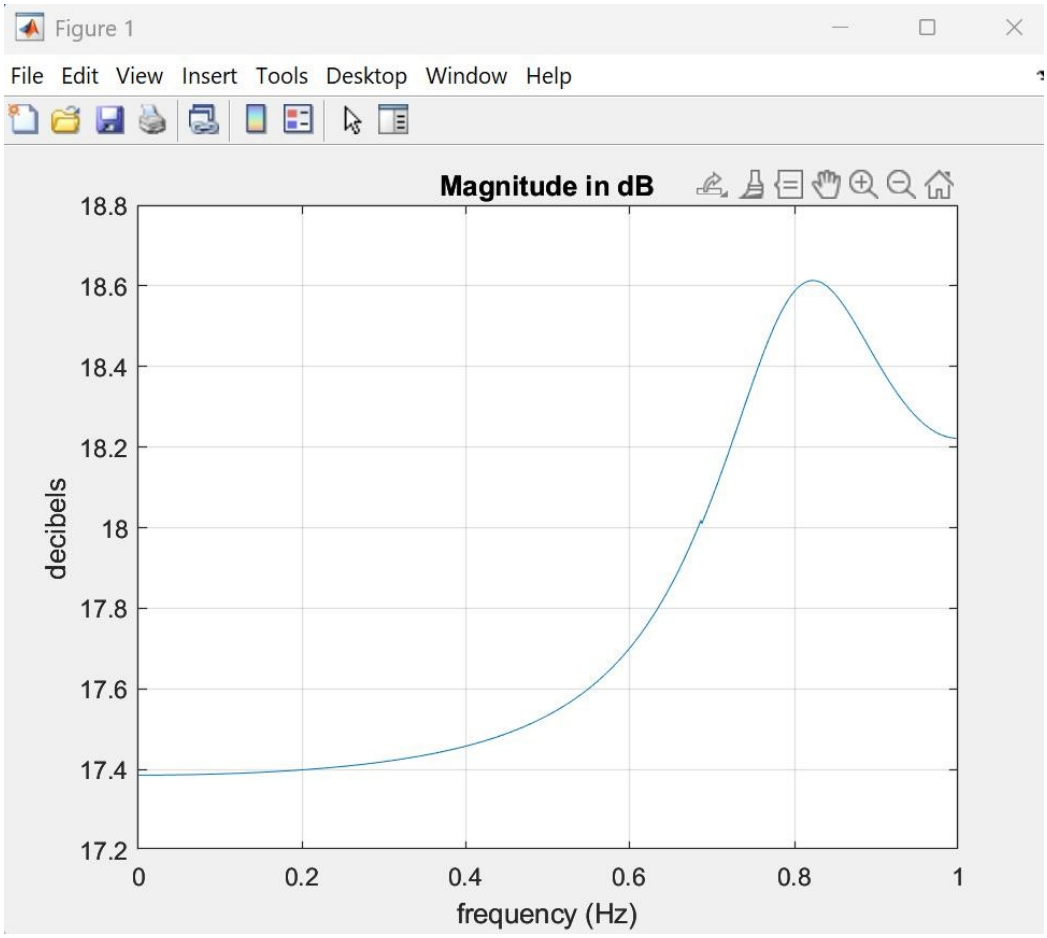
% Bilinear transformation:

```
clc;;
clear all;
close all;
wp = 0.75*pi; % digital passband frequency in Hz (normalized)
ws = 0.5*pi; % digital stopband frequency in Hz (normalized)
Rp=20*log10(1/sqrt(2)); % passband ripple in dB
Rp=-Rp;
As = 20*log10(0.2); % stopband attenuation in dB
As=-As;
% Analog Prototype Specifications:
Fs = 1;
 T = 1/Fs;
OmegaP = (2/T)*tan(wp/2); % prewarp prototype passband frequency
OmegaS = (2/T)*tan(ws/2); % prewarp prototype stopband frequency
% Analog butterworth Prototype Filter Calculation:
[n omegac]=cheb1ord(OmegaP,OmegaS,Rp,As,'s');
[c d] = cheby1(n,omegac,'high','s');
% Bilinear Transformation:
[b, a] = bilinear(c, d, Fs);
 [h db] = freqz(b,a);
plot(db/pi,20*log10(abs(h)));
axis([0 1 -20 1])
grid
xlabel('frequency (Hz)');
ylabel('decibels');
title('Magnitude in dB');
```


Figure 1
File Edit View Insert Tools Desktop Window Help
Magnitude in dB
decibels
0
-2
-4
-6
-8
-10
-12
-14
-16
-18
-20
0
0.2
0.4
0.6
0.8
1
frequency (Hz)

%IMPULSE INVARIENCE METHOD

```
clc;
clear all;
close all;
% digital passband frequency in Hz
(normalized)
wp = 0.75*pi;
% digital stopband frequency in Hz
(normalized)
ws = 0.5*pi;
% passband ripple in dB
Rp=20*log10(1/sqrt(2));
Rp=-Rp;
% stopband attenuation in dB
As = 20*log10(0.2);
As=-As;
% Analog Prototype Specifications:
Fs=1;
T = 1/Fs;
% prewarp prototype passband frequency
OmegaP = wp/T;
% prewarp prototype stopband frequency
OmegaS = ws/T;
% Analog butterworth Prototype Filter Calculation:
[n omegac] =cheb1ord (OmegaP, OmegaS, Rp, As, 's');
[c d] = cheby1 (n, omegac, 'high', 's');
% Impulse Invariant Transformation:
[b, a] = impinvar (c, d, Fs);
[h db] = freqz (b, a);
plot (db/pi, 20*log10(abs(h)));
grid;
xlabel ('frequency (Hz)');
ylabel ('decibels');
title ('Magnitude in dB');
```


Figure 1
File Edit View Insert Tools Desktop Window Help
Magnitude in dB
18.8
18.6
18.4
18.2
18
17.8
17.6
17.4
17.2
decibels
0
0.2
0.4
0.6
0.8
1
frequency (Hz)